

# Alejandro López

## Software Engineer/ Cloud Developer

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### ABOUT ME

I am a software engineer. I have been working since 2019 while finishing my engineering degree. I started working with mobile apps and then moved to create and maintain web applications as a full-stack developer. My goal is always to learn as much as possible and my main passion is system analysis and software development. I like to play the piano and produce electronic music.

### PROFESSIONAL EXPERIENCE

#### Hamal Solutions, Bogotá: Fullstack Dev / Fully Remote

Jul 2022 - Present

- **Implemented** and refined an **Agentic AI Chatbot** served through a **React** Web app with a **Flask** Backend. **Deployed** in **Azure** and connected to **Microsoft AI Search** service for SalesFactory AI to improve internal team insights and customer research.
- **Developed** an **AI-driven Web Application** using **Python** to handle customer service reviews from **Google** and **Yelp** for a customer to reduce the customer service team overload by 80%.
- **Developed** a management website in **AWS Amplify** for SalesFactory company to orchestrate marketing campaigns.
- **Maintained** and **improved E2E testing** strategies and test cases for multiple projects using **Cypress** reducing the bugs in production and improving test coverage to 95%.

#### Stipe, Guatemala: FullStack Android Dev / Fully Remote

Jun 2020 - Aug 2024

- Created and published a **Flutter Android/iOS** application to optimize oil palm production processes avoiding losses and improving productivity for 50+ users.
- Developed an **Express** backend connected to a postgres database deployed on an **EC2** instance for user data management and custom **JWT authentication**.
- Developed and deployed a **React** promotion Website with a **Github** Actions **CI/CD** to promote the mobile application.

### EDUCATION

UNET, Táchira: Computer Engineering

2016-2023

Informatic Engineer.

#### AWS Training and Certification: AWS Cloud Developer

2022-2022

AWS Skill Builder courses and certifications

### TECHNICAL SKILLS

**Cloud:** AWS (S3, EC2, DynamoDB, Amplify), Azure (Cosmos, Semantic Search, Container App, Azure Functions).

**Frontend:** React, Javascript, NodeJs, CSS, Tailwind CSS.

**Backend:** Node.js, Typescript, Express, Django, Flask.

**Gen AI:** Vector Databases, Semantic Search, Text Embedding, Langchain, Langgraph, RAG.

**Databases & Tools:** Postgres SQL, Sequelize, Git, Cypress, Jest, Flutter.

### LANGUAGES

Spanish-Native, English-C2, German-A1

## CERTIFICATIONS

**AWS Certified Cloud Developer** AWS Certification Exam

**AWS Certified AI Practitioner** AWS Certification Exam

**AWS Certified Cloud Practitioner** AWS Certification Exam

**Agentic AI** DeepLearning.ai

**Building AI Browser Agents** DeepLearning.ai

**Freecodecamp Back End Development and APIs**

**EF SET Certificate** EF SET Certificate 76/100 (C2 Proficient)

## PROJECTS

### Hamal Solutions Sales Factory AI FreddAid Web App: *Agentic AI project*

React, Flask, Azure, Microsoft AI Search, Typescript, Stripe

- **Engineered** an agentic AI chatbot using **Retrieval-Augmented Generation (RAG)** and **Python**, enabling automated data retrieval and code generation for diverse client services.
- **Architected** a sophisticated AI agent using **LangGraph** and **OpenAI**, deployed via **Azure Functions** for scalable, serverless execution.
- **Developed** a custom data ingestion pipeline using **Azure Cognitive Search** for indexing and embeddings generation to enhance retrieval accuracy.
- **Built** a full-stack web application with **React**, **TypeScript**, and **CosmosDB**, integrating **Microsoft B2C** for secure enterprise authentication.
- **Integrated Stripe** API to manage subscription logic and payment processing for commercial scalability.
- Implemented a **Model Context Protocol (MCP)** server to extend the chatbot's capabilities, currently serving as a critical internal tool for a 30-member team at Sales Factory AI.

### Hamal Solutions Sales Factory AI Kindness Quotient Web App: *AI Tool project*

Python, Quart, Azure, Typescript, React, Yelp API, Google business API, Google Console.

- **Engineered an AI-driven Review Management System** using **React** and **Quart (Python)**, automating the generation of context-aware responses for Google and Yelp reviews.
- **Integrated Google Business Profile API** and implemented **Web Scraping APIs** to aggregate customer feedback and star ratings across multiple platforms.
- **Implemented a Multi-tenant Architecture** designed to support scalable user onboarding and organizational data isolation for a SaaS-ready deployment.
- **Leveraged LLMs (OpenAI/similar)** to inject dynamic business promotions and deals into automated responses, enhancing marketing outreach through customer service.
- **Reduced customer service response time by 80%** for Sabor Restaurant by streamlining the review management workflow for an internal team of 15 members for 31 locations.
- **Architected the backend with Quart**, utilizing asynchronous Python to handle high-concurrency API requests and data processing efficiently.

### Hamal Solutions Sales Factory Orchestration App: *API/web app project*

React, AWS Amplify, Cognito, Material UI, Cypress, Jest, Postgres, Sequelize

- **Collaborated** with a cross-functional development team at Sales Factory to architect, build, and deploy a scalable web application utilizing **AWS Amplify Hosting**.
- **Developed** a robust backend API using **Express.js** and **Sequelize** ORM, efficiently managing complex data schemas and queries within **Google BigQuery**.
- **Engineered** a responsive and intuitive frontend using **React** and **Material UI (MUI)**, implementing a consistent design system and seamless user interface.
- **Integrated AWS Amplify** for secure user authentication, managing the end-to-end logic for login, signup, and session management.

- **Championed a Test-Driven Development (TDD)** approach, ensuring high code quality and reliability through automated testing with **Cypress (E2E)** for the frontend and **Jest** for the **Express API**.

### Hamal Solutions Progressive Label: *web app project*

Remix, Web app

- **Developed and implemented** a comprehensive suite of end-to-end (E2E) automated tests using **Cypress**, ensuring the stability of a **Remix** and **React** web application.
- **Optimized** backend reliability for a **Node.js** and **Express.js** infrastructure deployed on **AWS Elastic Beanstalk**.
- **Collaborated** with the engineering team to design and build full-stack features, including frontend views and backend endpoints for a centralized **Order Management System (OMS)**.
- **Engineered** custom modules for the **customer service portal** at Progressive Label Inc., improving internal workflows and data management.
- **Facilitated** seamless deployment and environment management within the **AWS ecosystem**, maintaining high availability for client-facing services.

### Stipe: *Guatemala startup*

Flutter Android iOS app, Web App

- **Spearheaded the legacy migration** of a native Android application from **Java** to **Flutter**, achieving a unified codebase and improved cross-platform performance.
- **Implemented complex statistical algorithms** to process user data across four oil palm production stages, providing automated optimization solutions for industrial workflows.
- **Developed robust data-handling features** allowing the application to parse **Excel (XLSX/CSV)** files and generate shareable performance reports.
- **Architected and deployed a Progressive Web App (PWA)** using **Flutter**, extending the reach of proprietary algorithms to web browsers.
- **Collaborated closely with UI/UX designers** to translate high-fidelity brand designs into a responsive and polished web interface for marketing and user acquisition.

### UNET Final Thesis Project: *UNET thesis*

React, Redux, Firebase, Django, Multiobjective evolutionary genetic algorithm Web App

- **Engineered a Multiobjective Evolutionary Genetic Algorithm** in **Python** as a terminal thesis project, optimizing travel itineraries based on complex user preference variables.
- **Developed a full-stack architecture** utilizing **Django** for the backend logic and **Firebase Firestore** as a NoSQL database for scalable hotel data storage.
- **Built a responsive frontend** with **React** and **Material UI**, implementing **Redux** for robust global state management and **Axios** for efficient asynchronous API communication.
- **Integrated diverse datasets** to perform multi-criteria analysis, resulting in an automated recommendation engine that balances user interests with logistical constraints.

### Hamal Solutions Little Bellies: *web app project*

React, Web app

- **Developed dynamic web interfaces** using **React** and **Node.js** for Pancita's Ultrasound, focusing on data-intensive views like summary tables and appointment dashboards.
- **Integrated multiple data sources**, including **AWS DynamoDB** for NoSQL storage and the **Mindbody API** for real-time appointment and scheduling management.
- **Ensured application reliability** by authoring comprehensive end-to-end (E2E) automated tests with **Cypress**, reducing manual QA overhead.

## HOBBY PROJECTS

**REACT-Simcity Build it**

Simcity simulation game in React with basic population functionality and stats control deployed in Vercel.

[React App \(simcity-buildit-guwil5o4i-alexies98.vercel.app\)](#)

**Three JS-Simple Solar System page**

HTML, CSS, and Javascript web page with some fun facts about the local solar systems using Three Js library and WebGL.

[First 3D example \(alexwebglfirstproject.vercel.app\)](#)

**ThreeJS-City Skyline**

React fiber, web project with some functions like (movements, traffic light, sounds and lights) with basic shapes.

<https://github.com/Alexjes98/i-walk-this-city>

**Three JS- Chess board**

HTML, CSS, and Javascript web page with a chess board with basic functions

<https://alex3-d-chess.vercel.app>

**Unity 3D LAN shooter tank basic game**

A 3D tank game in Unity with basic functionality and LAN configuration for multiplayer gameplay.

[Alexies98/AlexTankGame: a simple tank experiment for fun \(github.com\)](#)